

CONGENITAL ICHTHYOSIFORM ERYTHRODERMA

Congenital ichthyosiform erythroderma (CIE) is apparent at birth, with affected newborns typically presenting with a taut, shiny **collodion membrane**. After the membrane sheds, the skin remains diffusely red and is covered with a fine, white, generalized scale (see Fig. 47-5). On the lower legs, the scale may become larger and darker. In contrast to lamellar ichthyosis, classic CIE usually lacks significant **ectropion**, **eclabium**, or **alopecia**.

Sweating capacity varies widely; many infants and children with CIE have **minimal sweating** and exhibit **severe heat intolerance**. Mucous membranes are generally spared. Palmoplantar involvement is variable—some patients show mild thickening, while others have normal palms and soles. Nails may develop **ridging**, but are often unaffected. As with all forms of ichthyosis, there is an increased risk of **dermatophyte infections** of the skin and nails.

Histopathologically, CIE is characterized by **hyperkeratosis**, **acanthosis**, and frequent **parakeratosis** (see Fig. 47-2C). Studies have demonstrated **markedly increased epidermal cell turnover**, which contrasts with the slower turnover seen in lamellar ichthyosis.[49]

Genetically, CIE is part of the autosomal recessive congenital ichthyosis (ARCI) spectrum. A small subset of patients with CIE have mutations in the **\$TGM1\$** gene. Additionally, two siblings with a mixed phenotype—CIE and palmoplantar keratoderma—were found to have an autosomal dominant mutation in **loricrin**, a cornified envelope protein.[54] Genetic heterogeneity in ARCI is well established: **\$TGM1\$** has been excluded as the causative gene in some families with classic lamellar ichthyosis or CIE phenotypes, and mutations in other genes have been identified.

To date, mutations in **five additional genes** have been linked to nonsyndromic ARCI, and **two more** to syndromic forms. Two genetic loci—**12p11.2-q13** and **19p13.1-13.2**—are associated with ARCI, though the responsible genes remain unidentified (Table 47-3). A newly proposed **lipxygenase pathway** may serve as a unifying mechanism linking many forms of ARCI.[20] However, this hypothesis requires further validation, and the connection between the most common ARCI mutation (**\$TGM1\$**) and lipid abnormalities remains unclear.

EPIDERMOLYTIC HYPERKERATOSIS

In 1902, Brocq described **bullous ichthyotic erythroderma**, distinguishing the blistering form from nonblistering congenital ichthyotic erythroderma.[55] His report included three unrelated patients with variable clinical features, likely representing the first documented cases of what is now known as **epidermolytic hyperkeratosis** (OMIM #113800).

The condition is named for its hallmark histopathologic findings: **vacuolar degeneration** of the epidermis (epidermal lysis) and **hyperkeratosis**. It is also referred to as **bullous congenital ichthyosiform erythroderma**, a descriptive term highlighting the neonatal onset of blistering, erythema, scaling, and erosions.

Epidermolytic hyperkeratosis follows an **autosomal dominant** inheritance pattern, with an estimated prevalence of **1 in 200,000 to 300,000** individuals. However, **spontaneous mutations** are common, accounting for up to **50% of cases**, which therefore lack a family history.

Clinically, the disease presents at birth with **widespread blistering, erythema, and skin peeling** (Fig. 47-6).